

Summary of Chapter 2 and Detailed Discussion of Tumour Radiotherapy

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1 Summary of Chapter 2

Chapter 2 of the *Radiation Biology: A Handbook for Teachers and Students* provides the scientific foundation of radiation biology by introducing the physical principles of ionizing radiation, its interactions with matter, molecular and cellular mechanisms of radiation damage, and the biological principles underlying radiotherapy and radiation protection.

The chapter is organized into five major sections.

1.1 Physics and Chemistry of Radiation Interactions with Matter

This section discusses:

- Sources of ionizing radiation
- Electromagnetic and particulate radiation
- Photon interactions (photoelectric effect, Compton scattering, pair production)
- Linear Energy Transfer (LET)
- Radiation dose quantities
- Radiation dosimetry
- Direct and indirect radiation effects

The chapter emphasizes that biological damage begins with energy deposition through ionization. Radiation may damage biomolecules directly or indirectly through radiolysis of water, producing reactive free radicals.

1.2 Molecular and Cellular Radiobiology

This section explains:

- DNA damage mechanisms
- DNA repair pathways
- Cell cycle effects
- Cell survival curves
- Relative Biological Effectiveness (RBE)

- Radiation sensitizers
- Radioprotectors
- Bystander effects

DNA double-strand breaks are identified as the most critical lesions responsible for radiation-induced cell death.

1.3 Tumour Radiotherapy

Major topics include

- Tumour growth
- Tumour response to radiation
- Tumour control probability
- Dose fractionation
- Tumour hypoxia

These concepts form the biological basis of modern radiation therapy.

1.4 Normal Tissue Response to Radiotherapy

Topics include

- Acute tissue injury
- Late tissue injury
- Tissue repair
- Therapeutic ratio
- Whole-body irradiation

The objective is to maximize tumour control while minimizing damage to healthy tissues.

1.5 Radiobiological Basis of Radiation Protection

This section discusses

- Radiation accidents
- Radiation-induced cancer
- Embryonic and fetal effects
- Heritable genetic effects
- Epidemiological evidence for radiation risks

These concepts form the scientific basis for radiation protection standards and dose limits.

2 Detailed Discussion of Section 2.4: Tumour Radiotherapy

Section 2.4 is one of the most important parts of radiation biology because it combines radiation physics, molecular biology, tumour biology, and clinical oncology. It explains why ionizing radiation can selectively destroy tumour cells while preserving surrounding healthy tissues.

2.1 Tumour Growth

Tumour growth depends on cellular proliferation, cell loss, angiogenesis, nutrient availability, and oxygen supply.

For small tumour populations, exponential growth may be approximated by

$$N(t) = N_0 e^{\lambda t}, \tag{1}$$

where

- N_0 is the initial number of tumour cells,
- λ is the growth constant,
- $N(t)$ is the tumour population after time t .

Most clinical tumours do not continue exponential growth indefinitely because nutrient and oxygen limitations reduce proliferation. Instead, tumour growth is better represented by the Gompertz model,

$$N(t) = N_0 \exp \left[\frac{G}{B} (1 - e^{-Bt}) \right], \quad (2)$$

where

- G represents the initial growth rate,
- B describes progressive slowing of growth.

This slowing results from poor vascularization, hypoxia, nutrient depletion, and waste accumulation.

2.2 Tumour Response to Irradiation

Ionizing radiation deposits energy into tumour cells through ionization and excitation.

The most lethal lesions are DNA double-strand breaks (DSBs).

Cell survival following irradiation is described by the Linear–Quadratic (LQ) model,

$$S(D) = e^{-(\alpha D + \beta D^2)}, \quad (3)$$

where

- $S(D)$ is the surviving fraction,
- D is the absorbed dose,
- α represents single-track damage,
- β represents double-track interactions.

At low doses, the linear term dominates, whereas at higher doses the quadratic term becomes increasingly important.

The ratio

$$\frac{\alpha}{\beta}$$

characterizes tissue sensitivity to dose fractionation.

Typical values are

- Tumours and acute tissues:

$$\alpha/\beta \approx 10 \text{ Gy}$$

- Late-responding tissues:

$$\alpha/\beta \approx 2 - 3 \text{ Gy}$$

2.3 Tumour Control Probability (TCP)

Radiotherapy aims to eliminate every clonogenic tumour cell.

If

- N is the number of clonogenic cells,
- $S(D)$ is the surviving fraction,

then the expected number of surviving clonogens is

$$N_s = NS(D). \tag{4}$$

Assuming a Poisson distribution,

$$\boxed{TCP = e^{-NS(D)}} \tag{5}$$

where TCP denotes Tumour Control Probability.

Larger tumours require higher doses because they initially contain more clonogenic cells.

2.4 Dose Fractionation

Instead of administering the total radiation dose in one session, treatment is divided into multiple fractions.

Fractionation exploits the Five Rs of Radiobiology:

1. Repair
2. Redistribution
3. Reoxygenation
4. Repopulation
5. Radiosensitivity

These biological processes improve tumour control while reducing normal tissue toxicity. The Biological Effective Dose (BED) is

$$BED = nd \left(1 + \frac{d}{\alpha/\beta} \right) \quad (6)$$

where

- n is the number of fractions,
- d is the dose per fraction.

BED enables comparison between different treatment schedules.

2.5 Tumour Hypoxia

Tumour hypoxia is one of the major causes of radioresistance.

Oxygen stabilizes radiation-induced DNA damage by reacting with free radicals generated during irradiation.

The Oxygen Enhancement Ratio (OER) is defined as

$$OER = \frac{D_{\text{hypoxic}}}{D_{\text{oxygenated}}} \quad (7)$$

where

- D_{hypoxic} is the dose required under hypoxic conditions,
- $D_{\text{oxygenated}}$ is the dose required under oxygenated conditions.

For low-LET radiation,

$$OER \approx 2.5 - 3.$$

Therefore, hypoxic tumour cells may require nearly three times the radiation dose to achieve the same biological effect.

Methods to overcome tumour hypoxia include

- Hyperbaric oxygen therapy
- Carbogen breathing
- Hypoxic cell radiosensitizers
- High-LET particle therapy

2.6 Clinical Importance

Tumour radiotherapy is fundamentally a balance between maximizing tumour eradication and minimizing injury to surrounding normal tissues.

Modern treatment planning combines

- the Linear–Quadratic model,
- Tumour Control Probability,
- Biological Effective Dose,
- Oxygen Enhancement Ratio,
- fractionation principles,
- tumour hypoxia,
- and clonogenic cell survival

to optimize treatment outcomes.

3 References

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